

Homework #1 (Due: April 1st, during class hour)

Answer the following questions. Write your answers on a paper and submit during the class hour.

QUESTIONS

1. Write a BNF description of the Boolean expressions of C, including the three operators &&, ||, and ! and the relational expressions.
2. Consider the following grammar:

$$\langle S \rangle \rightarrow \langle A \rangle \langle B \rangle \langle C \rangle$$
$$\langle A \rangle \rightarrow a \langle A \rangle \mid a$$
$$\langle B \rangle \rightarrow b \langle B \rangle \mid b$$
$$\langle C \rangle \rightarrow c \langle C \rangle \mid c$$

Show the parse tree and leftmost derivation for the string “aaabbbbcc”.

3. Prove that the following grammar is ambiguous.

$$\langle S \rangle \rightarrow \text{begin } \langle A \rangle \text{ end}$$
$$\langle A \rangle \rightarrow \langle A \rangle \wedge \langle A \rangle \mid \langle A \rangle + \langle A \rangle \mid \langle \text{id} \rangle$$
$$\langle \text{id} \rangle \rightarrow x \mid y \mid z$$

4. Compute the weakest precondition for the below program segment:

if ($x > y$)

$b = x$;

else

$b = y$;

{ $b > 0$ }

5. Show a complete parse, including the parse stack contents, input string, and action for the string **(id + id) * (id + id)\$**, using the grammar and parse table in Section 4.5.3